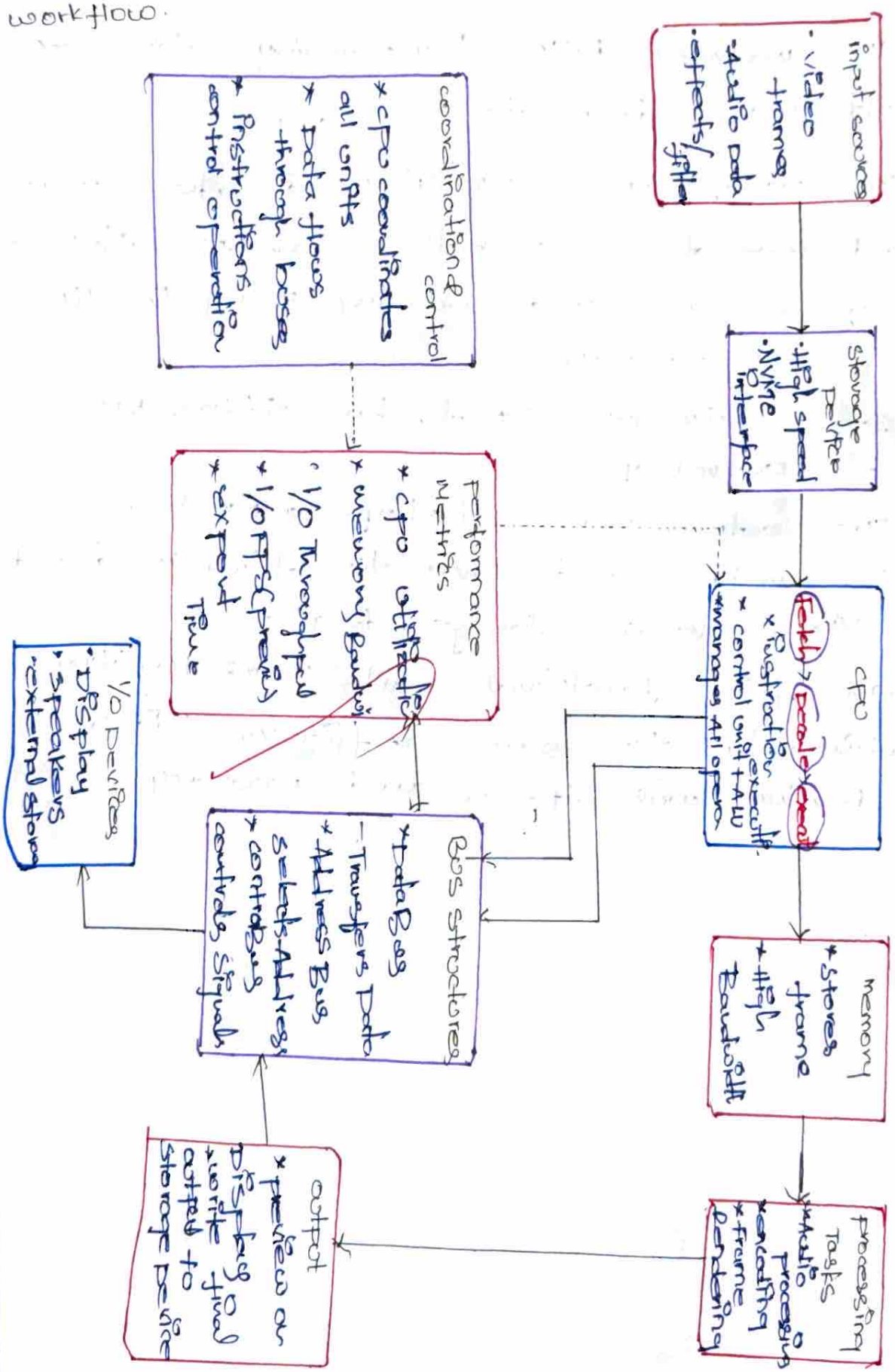


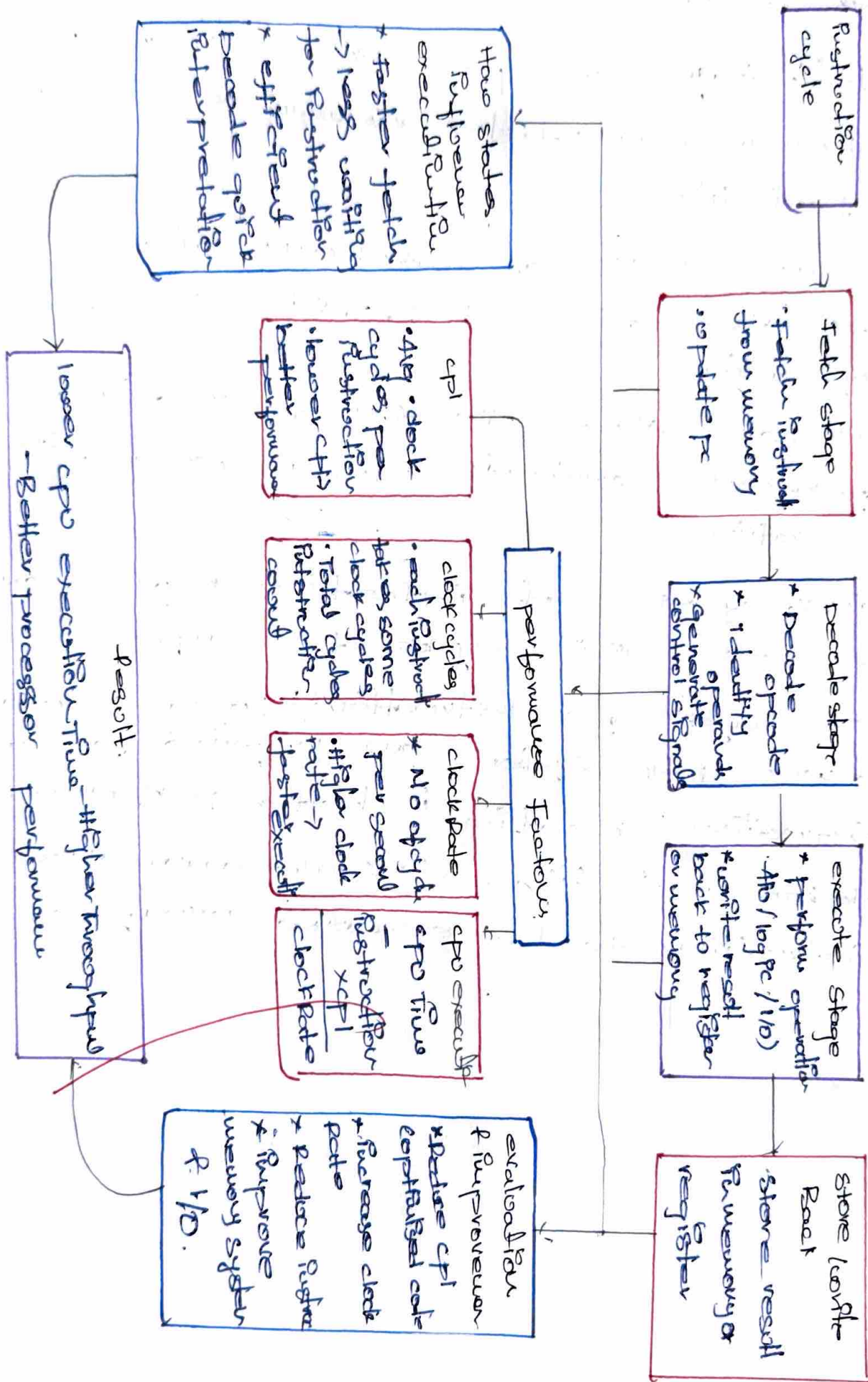
Q1. concept Map - 4K VIDEO editing and exporting workflow.



sub question.

- * Input unit accepts video frames and audio data from the SSP
- * The memory (RAM) temporarily stores the multimedia data for fast access.
- * The CPU fetches instructions, decodes them and executes operation such as filtering, compression and encoding using the ALU and control unit.
- * Bus structure (data bus, Address Bus, and control Bus).
- * The output unit displays a preview on the monitor and saves the final rendered 4K video to the storage device.
- * All five functional units work together continuously to ensure efficient video rendering with speed and accuracy.

concept map - instruction execution cycle and performance



Q sub-question

Fetch stage: The CPU retrieves instructions from memory. faster memory access reduces fetch time.

Decode stage: The control unit interprets the instruction and identifies the required operation.

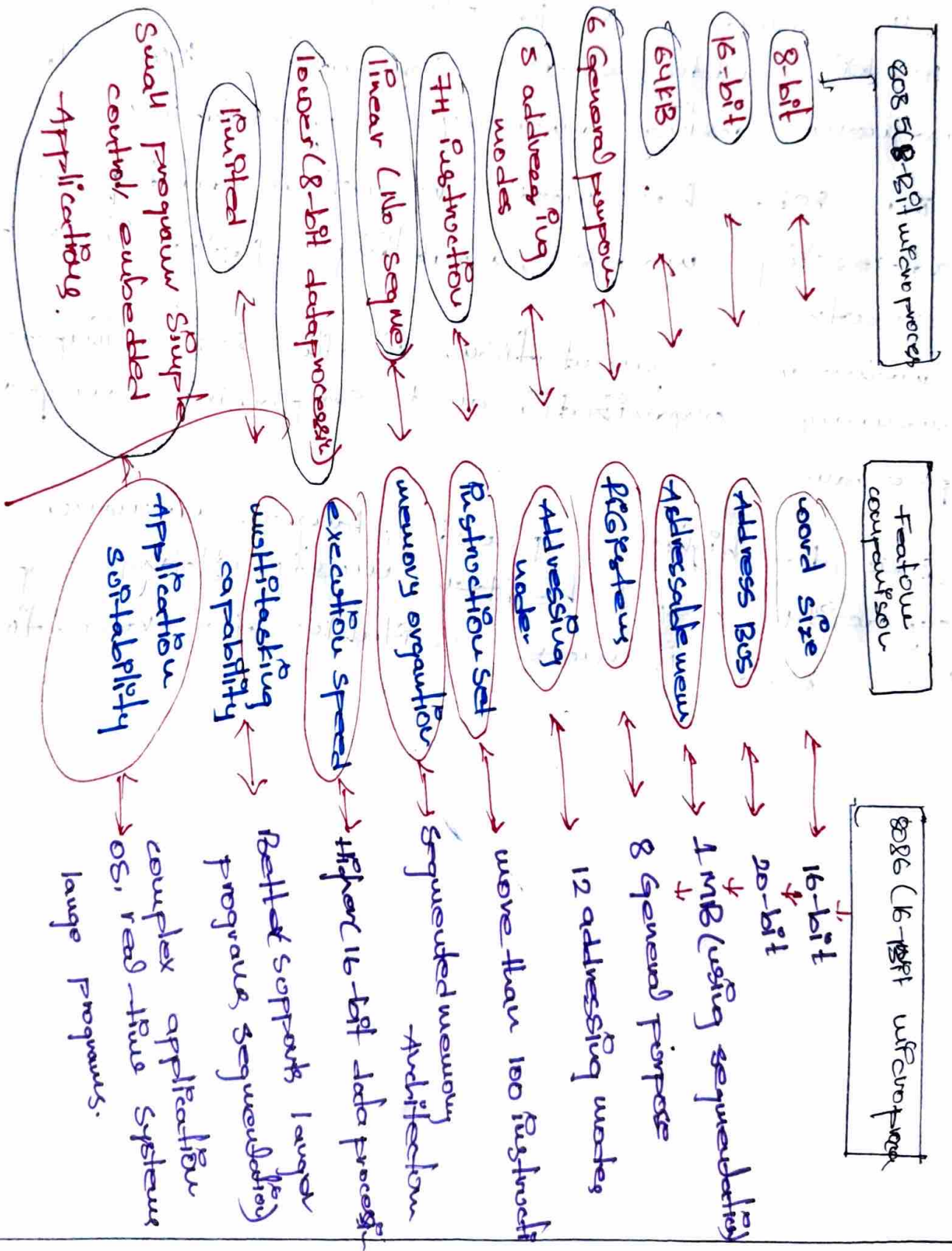
execute stage: The ALU or execution unit performs arithmetic, logical, or data transfer operations.

clock cycles: each instruction requires one or more clock cycles to complete.

CPI (Cycles per instruction): Lower CPI means fewer clock cycles are required for each instruction, resulting in better performance.

clockrate: A higher clock rate allows more instructions to be executed every second.

concept map - comparison of 8085 and 8086 Architecture

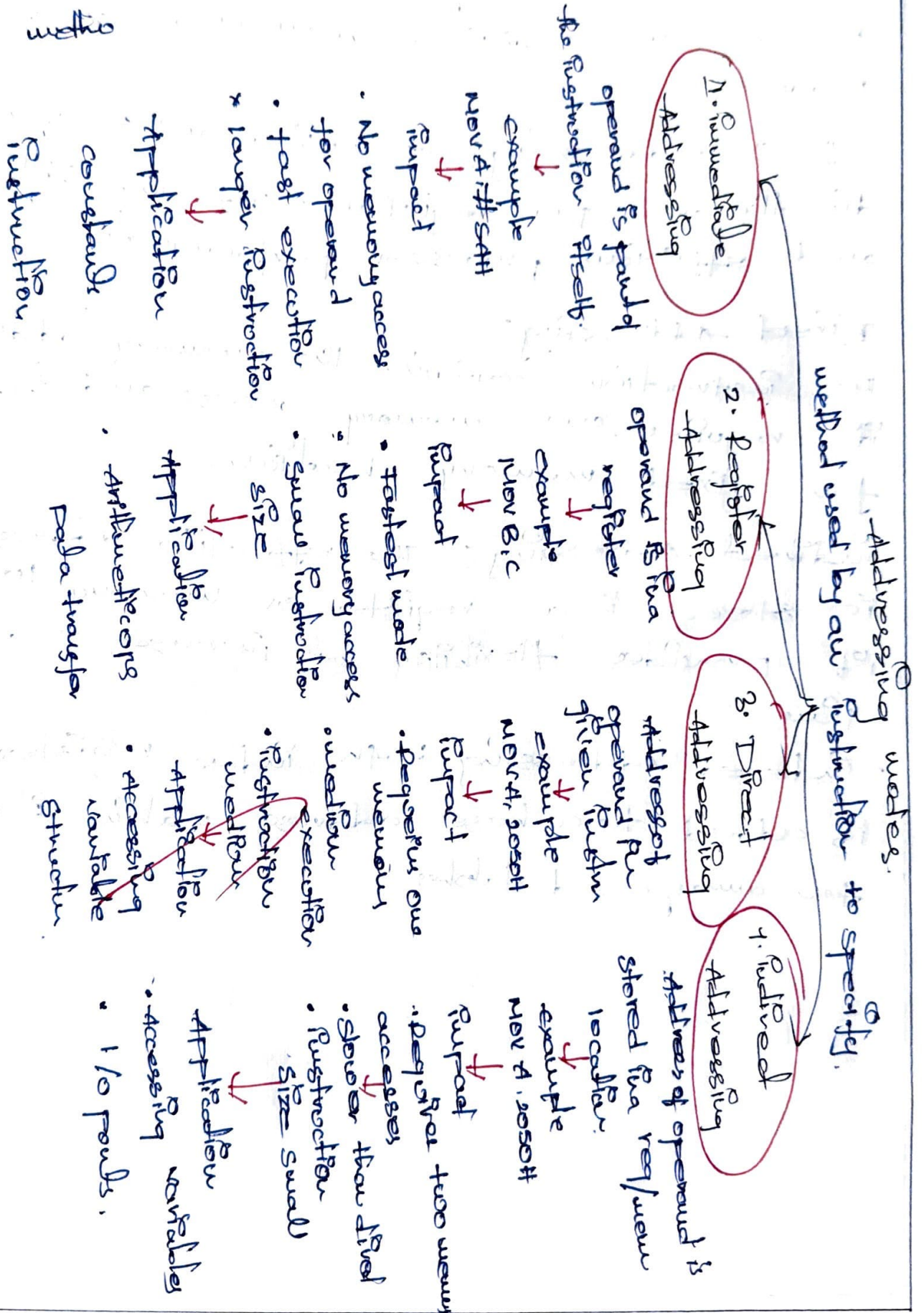


sub question:

- * The 8085 is an 8-bit processor with a 16-bit address bus, capable of addressing 64 KB of memory.
- * The 8086 is a 16-bit processor with a 20-bit address bus, supporting MIB of memory using memory segmentation.
- * The 8086 has more registers and powerful addressing modes, enabling faster instruction execution.
- * memory segmentation in the 8086 improves memory organization and supports larger programs.
- * Due to higher speed, larger memory capacity, and better multitasking support the 8086 is more suitable for real-time.

concept map. - Addressing modes and their impact

Addressing modes



Sub question

Immediate Addressing: Data is stored within the instruction. It provides the fastest execution because no additional memory access is required.

Register Addressing: operations are performed directly on CPU registers, offering high speed and efficient processor performance.

Direct Addressing:

The instruction contains the memory address it requires one memory access and is suitable for fixed memory location.

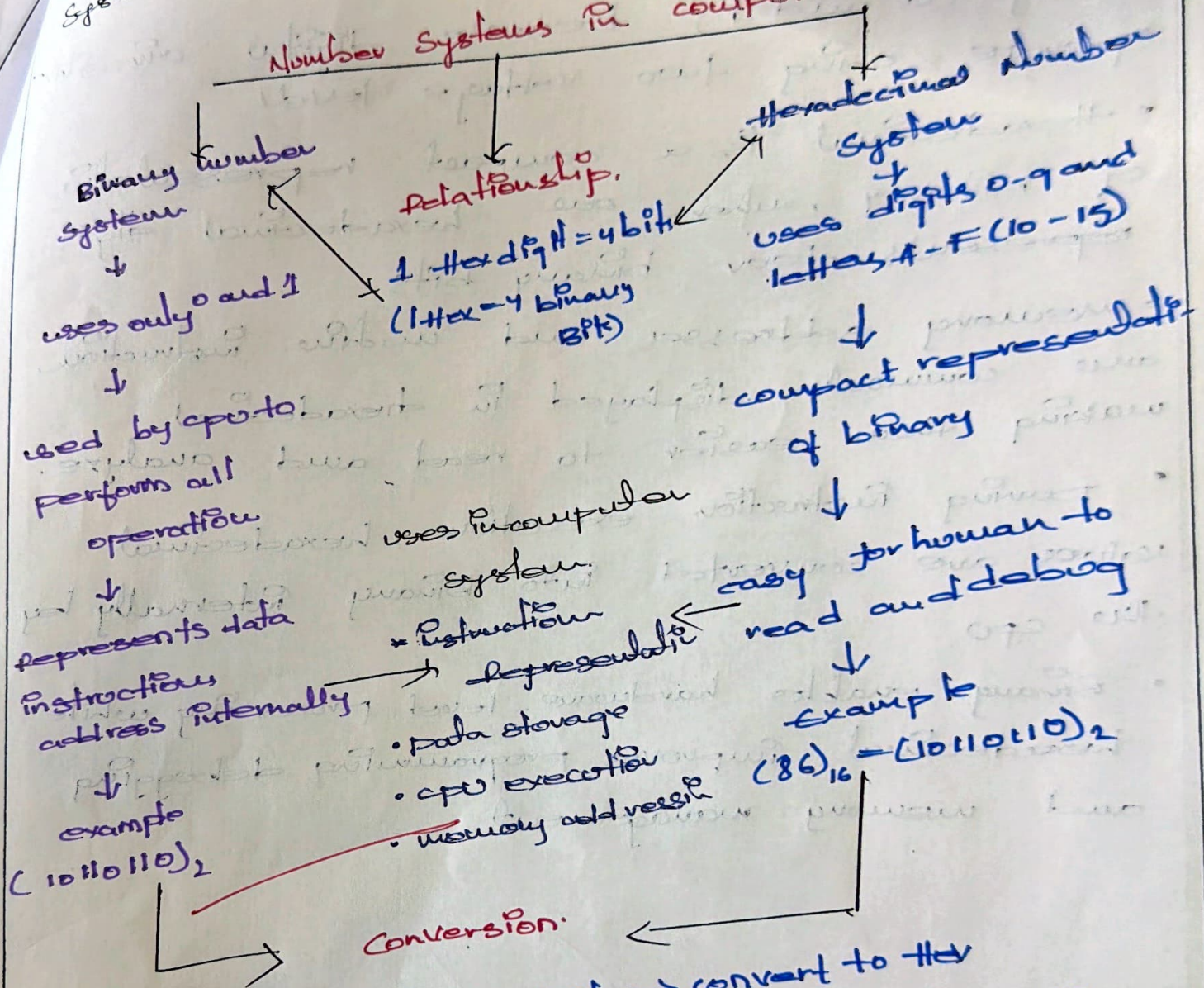
Indirect Addressing: The effective address is stored in a register or memory location. It provides flexibility but increases execution time.

Indexed Addressing: An index register is added to a base address, making it ideal for arrays and tables.



concepts - Binary and hexadecimal number systems

Number Systems in computer system.



Binary $\rightarrow$ Group 4 bits $\rightarrow$ convert to hex

Hex $\rightarrow$ convert each hex digit $\rightarrow$ 4 bits binary

sub question

- The CPU stores and executes instructions in binary (0 and 1) because digital circuits operate using two voltage levels.
- Hexadecimal is a compact representation of binary, where one hexadecimal digit represents four binary bits.
- Memory addresses and machine instructions are commonly displayed in hexadecimal, making them easier to read and analyze.
- During instruction execution, hexadecimal values are converted into binary internally by the CPU.
- Binary enables hardware-level processing while hexadecimal improves programming debugging and memory management.

